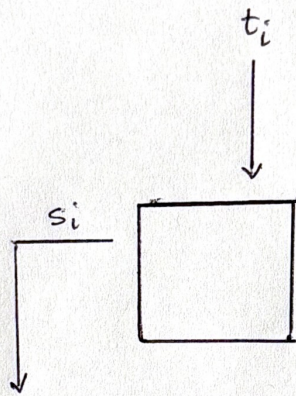


Appendix - Noken²

Noken Network Connections: Composition and Feedback



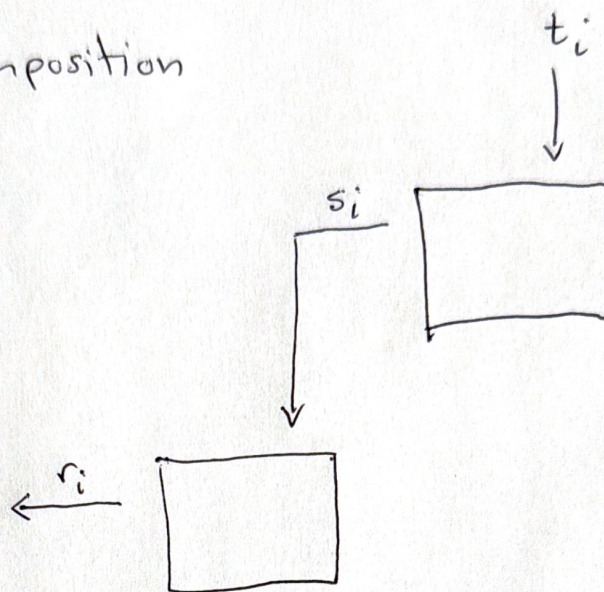
denotes:

$$s_i = \varphi(Vs_i + Qt_i)$$

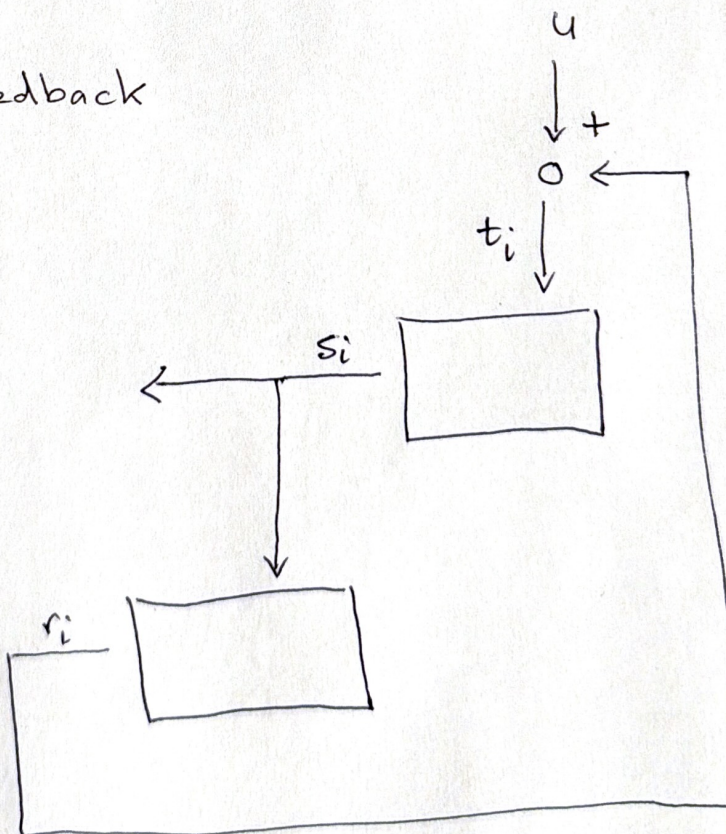
$$\varepsilon = {}^sKs_i + {}^tKt_i$$

where V is well-posed for φ and $\varepsilon \in \mathbb{R}_+^n$.

Cascade - composition



Synthesis - feedback



$$t_i = u + r_i$$

In Zen, thought is arrived at by a cascading process. Nen, axiomatic agents, receive input either from the environment or other Nen including themselves. The input is then both reflected and synthesized, creating two outputs to the environment. 'Receive, reflect, synthesize.' Manifestation of thought is this cascade of Nen interactions. We now utilize this paradigm in Noken networks.

Think of an agent as being a kind of implicit deep learning model [5]. The agent receives an input token from a context string and outputs a Noken. Note that besides being composed, implicit models can be joined into feedback loops as follows: agent A receives input from the environment plus output from agent B. The output of A goes to the environment, as well as being input to B, thereby creating a feedback loop synthesizing information. So in this way, agents just like Nen, receive, reflect and synthesize tokens. Interpret such cascades of agent interactions as machine thought. Thought is directed by the architecture of the agent network much like the human brain. In summary, a stimulus, a token from a context string, generates a network cascade of receiving, reflecting and synthesis, as Noken agents interact through composition and feedback. The opportunity to interact being dictated by proximity in a network graph.

So why don't these cascades go on forever? Friston [6] posits that biological agents minimize free energy. In this case, optimization is halted when the energy required for exploration exceeds that being exploited. Similarly, Noken implicit model outputs are refined while their Renyi α -entropy continues to be reduced, halting when this doesn't occur, or is acceptably small, or becomes too expensive to learn. These thresholds are hyperparameters that limit cascading.

Networks of Noken agents could be trained on domain specific corpuses of data. These subject matter experts of specialized Nokens can then be used to build a knowledge graph sharing their collective experience, learning from one another using an internal universal interlingua. In this way, cross disciplinary analogies are generated from multiple thought paradigms, enabling heuristics to look at matters from a fresh perspective.

Feedback loops in implicit models play a crucial role in enhancing their performance. These loops allow the model to adjust its outputs based on previous results, leading to improved generalization and adaptability.

The key aspects of feedback and composition in these models include:

- Closed-Loop Feedback: This mechanism enables the model to refine its predictions by continuously integrating new information.
- Depth Adaptability: Implicit models can adjust their complexity based on the task, which helps in managing varying data distributions.

Cascades of interactions in a knowledge graph, by Noken agents using composition and feedback loops, is a blueprint for emergent machine thinking.

This begs the question, how do we configure knowledge graphs, train them and raise self learning entities to be useful virtual citizens? It is easy to fall back into anthropomorphic paradigms when contemplating the ramifications of thinking Noken agents. There are parallels but also significant differences between people and Nokens. For one, human input is through our senses and our purpose to propagate our genes whereas Noken inputs are tokens or Noken outputs, whose education, training and possible emergent behavior remains to be explored. Do they need purpose while making discoveries? What utility will we and 'they' have for each other? In the narrowest sense, Noken agents enable semantic reasoning when learning in next-token and sequence-to-sequence applications but being able to meaningfully converse with them portends radical redefinition of artificial intelligence and our intercourse. Beyond philosophical debate, Noken agents can be implemented and empirically tested. Ironically, just as we don't know what another person is thinking, how will we study 'homo nok machina'? At least with Nokens we can track the evolution of their keys that bridge the meaning of queries and values.

A genesis for further inquiry:

- An implicit model [1] tells us which query, key, value matrices to use and how they're related
- Language semantics is captured by the key-query and key-value pairs, so their training is crucial to fidelity of meaning
- The implicit model fixed point is a latent source token, that we call a **Noken**
- Noken variations [2] demonstrate there are multiple ways to update queries besides transformer attention
- Notes [1] and [2] give entropy criteria for refining queries
- Attention dynamics note [3] tells us when refinement converges

Key Noken features to develop:

- Keys bridge the semantics of queries and values
- Similarity functions align meaning
- Entropy of similarity probabilities, measures the degree of semantic fidelity
- Query refinement improves semantic alignment
- Next-token and seq-to-seq learning are semantic handshakes facilitated by a noken universal machine interlingua
- Beyond optimization problems, Nokens can be composed in chains [4] and configured into feedback loops enabling information cascading and synthesis. This self learning ability in turn generates a virtuous spiral of Noken universal machine interlingua expression with deepening semantics and reasoning.

Upshot: Noken are a blueprint for thinking machines.

Transformer attention, underlying LLM architecture, is the statistical tip of a latent semantic iceberg. Nokens tell us when and why a LLM will work, facilitating transition from being stochastic parrots to being semantic reasoners.

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